

Stochastic epidemic models with inference

Exercise Session 1

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Exercise 1.1

Final Size of Outbreak

Final Size Equation

$$1 - \tau = e^{-R_0\tau}.$$

Note: This equation only gives the final fraction infected among the **initially susceptible** individuals.

- There is always a solution $\tau = 0$.
- If $R_0 > 1$, there exists a second solution $\tau^* > 0$.
- Final size τ shall be the largest solution on $[0, 1]$.

Plot of final size as a function of R_0

Procedure in R:

- Set a function of R_0 solving for τ numerically, return $\tau = \tau(R_0)$.
- Create a vector of 10000 R_0 values in $[0, 5]$.
- Create a vector of corresponding values of τ .
- Plot τ against R_0 .

Now assuming initial fraction of immune

If there is a fraction r of **initially immune**, then there is fraction $(1 - r)$ of initially susceptibles. Then the final size among **initially susceptibles** τ^* solves

$$1 - \tau = e^{-R_0(1-r)\tau}.$$

Then the **overall** fraction infected shall be $\tau^*(1 - r)$.

Exercise 1.2

Markovian SIR Epidemic Model

- Consider a closed and homogeneous mixing population with fixed size N .
- At any time point, each individual is susceptible, infectious or recovered.
- The rate of infectious contacts is β , so the rate at which one infectious has contact with a specific other individual is β/N .
- Once infected, one remains to be infectious for a period $I \sim \text{Exp}(\gamma)$, after which one becomes recovered and immune.

Markovian SIR Epidemic Model

Let $S(t), I(t), R(t)$ be the number of individuals in different states S, I, R at time t respectively.

Two types of events:

- $S \rightarrow I$: a susceptible gets infected.
- $I \rightarrow R$: a infectious individual recovers.

The corresponding rates:

- $\frac{\beta}{N} S(t) I(t)$
- $\gamma I(t)$

Note: at any time t , $S(t) + I(t) + R(t) = N$.

Deterministic SIR Epidemic Model

SIR differential equation system

$$\begin{aligned}\frac{dS(t)}{dt} &= -\frac{\beta}{N}S(t)I(t), \\ \frac{dI(t)}{dt} &= \frac{\beta}{N}S(t)I(t) - \gamma I(t).\end{aligned}$$

Initial Conditions

$$S(0) = N - 1, I(0) = 1.$$

Numerical Solution of the SIR ODE

- Define the function to compute derivatives ($dS(t)/dt, dI(t)/dt$) for the SIR ODE.

```
1 gamma <- 0.25
2 beta <- 0.75
3 deter$_sir <- function(t,y, parms) {
4   beta <- parms[1]
5   gamma <- parms[2]
6   N <- parms[3]
7   S <- y[1]
8   I <- y[2]
9   return(list(c(S=...,
10               I=...,
11               )))
12 }
```

Numerical Solution of the SIR ODE

- Solve the SIR differential equation system with initial conditions (Use `deSolve::lsoda`):

```
1 lsoda(y= ..., \#initial conditions
2 times= ....., \#times at which explicit estimates
   for y are desired
3 func= ..., \#an R-function that computes the
   values of derivatives in the ODE
4 parms= ... \#vector or list of parameters used
   in func
5 )
```

Stochastic SIR Epidemic Model

Described as a continuous-time Markov process:

Events	Transition	Rates
Infection	$(S(t), I(t)) \rightarrow (S(t) - 1, I(t) + 1)$	$\frac{\beta}{N} S(t) I(t)$
Recovery	$(S(t), I(t)) \rightarrow (S(t), I(t) - 1)$	$\gamma I(t)$

Once $I(t) = 0$, the epidemic stops.

Algorithm to decide which event occurs first:

- From those two rates, we draw two exponential random numbers for each possible event. `rexp(...)`
- Determine the event with the smaller random number. `which.min(...)`
- Record the event time and update the number of S and I according to the event type.

Exercise 1.3

Markovian SEIR Epidemic Model

- Consider a closed and homogeneous mixing population with fixed size N .
- The rate of infectious contacts is β , so the rate at which one infectious has contact with a specific other individual is β/N .
- Individuals that get infected are first latent(exposed) for a exponentially distributed period with mean $1/\rho$, then they become infectious for a random duration $I \sim \text{Exp}(\gamma)$, after which they become recovered and immune.

Markovian SEIR Epidemic Model

Let $S(t), E(t), I(t), R(t)$ be the number of individuals in four states S, E, I, R at time t respectively. Assume that $S(0) = N - 1, E(0) = 0, I(0) = 1, R(0) = 1$.

Three types of events:

- $S \rightarrow E$: a susceptible gets infected.
- $E \rightarrow I$: an exposed individual become infectious.
- $I \rightarrow R$: a infectious individual recovers.

The corresponding rates:

Markovian SEIR Epidemic Model

Let $S(t), E(t), I(t), R(t)$ be the number of individuals in four states S, E, I, R at time t respectively. Assume that $S(0) = N - 1, E(0) = 0, I(0) = 1, R(0) = 1$.

Three types of events:

- $S \rightarrow E$: a susceptible gets infected.
- $E \rightarrow I$: an exposed individual become infectious.
- $I \rightarrow R$: a infectious individual recovers.

The corresponding rates:

- $\frac{\beta}{N}S(t)I(t)$
- $\rho E(t)$
- $\gamma I(t)$

Deterministic SEIR Epidemic Model

SEIR differential equation system

$$\begin{aligned}\frac{\partial S(t)}{\partial t} &= -\frac{\beta}{N}S(t)I(t), \\ \frac{\partial E(t)}{\partial t} &= \frac{\beta}{N}S(t)I(t) - \rho E(t), \\ \frac{\partial I(t)}{\partial t} &= \rho E(t) - \gamma I(t).\end{aligned}$$

Initial Conditions

$$S(0) = N - 1, E(0) = 0, I(0) = 1, R(0) = 0.$$